

NMTF-DTI: A Nonnegative Matrix Tri-factorization Approach With Multiple Kernel Fusion for Drug-Target Interaction Prediction

Ali Akbar Jamali , Anthony Kusalik , and Fang-Xiang Wu 

Abstract—Prediction of drug-target interactions (DTIs) plays a significant role in drug development and drug discovery. Although this task requires a large investment in terms of time and cost, especially when it is performed experimentally, the results are not necessarily significant. Computational DTI prediction is a shortcut to reduce the risks of experimental methods. In this study, we propose an effective approach of nonnegative matrix tri-factorization, referred to as NMTF-DTI, to predict the interaction scores between drugs and targets. NMTF-DTI utilizes multiple kernels (similarity measures) for drugs and targets and Laplacian regularization to boost the prediction performance. The performance of NMTF-DTI is evaluated via cross-validation and is compared with existing DTI prediction methods in terms of the area under the receiver operating characteristic (ROC) curve (AUC) and the area under the precision and recall curve (AUPR). We evaluate our method on four gold standard datasets, comparing to other state-of-the-art methods. Cross-validation and a separate, manually created dataset are used to set parameters. The results show that NMTF-DTI outperforms other competing methods. Moreover, the results of a case study also confirm the superiority of NMTF-DTI.

Index Terms—Drug-target interaction, target similarity, drug similarity, nonnegative matrix tri-factorization

1 INTRODUCTION

THE prediction of drug-target interactions (DTIs) is an essential step in drug discovery. Both experimental and computational approaches are applied to predict DTIs. Although experimental methods provide more precise results, they require a wide range of equipment, large financial resources, and substantial time investment — factors that hinder their applications. In contrast, computational methods tend to avoid these limitations. The timeliness, flexibility, and capacity for the high throughput of computational methods make them favorable for the DTI prediction. In addition, computational methods are attractive as they provide complementary alternatives to experimental approaches. Computational approaches to the DTI prediction have been facilitated by the large amount of biomedical data in public databases [1], [2], [3].

Many different computational approaches have been developed to predict DTIs. These methods vary according to their focus and algorithmic basis. The focus could be on targets [4], ligands [5], [6], or drugs [7]. Possible algorithmic bases include machine-learning models [1], [8], [9], [10], [11], [12], [13], graphs and networks [1], [14], [15], [16] and text mining, [17], [18], [19], [20], [21]. Most methods integrate heterogeneous information on targets, drugs, and drug-target interactions.

Machine learning approaches use positive and negative samples with classifiers—including support vector machine (SVM), K-nearest neighbor (KNN), and random forest (RF)—to predict interactions between drugs and targets. Graph-based methods such as $L_{2,1}$ -GRMF [22], NRLMF [23], NeoDTI [1], DDR [3] use the heterogeneous structure of drug-target interaction networks to predict DTIs. Recent studies [1], [10], [21], [24], [25] have shown that the applications of different heterogeneous data sources can significantly increase the performance of DTI prediction. For instance, Ding *et al.* [26] develop a multi-kernel method to predict DTIs, where different similarity matrices of drugs and targets are integrated. Their results show that integrating various target-related and drug-related data sources is an effective strategy to improve prediction performance in terms of AUC (area under the receiver operating characteristic curve) and AUPR (area under the precision and recall curve). Liu *et al.* [23] propose a neighborhood regularized logistic matrix factorization (NRLMF) method to predict unknown DTIs. NRLMF identifies the probability of an interaction between a drug and target via a logistic function of drug- and target-specific latent vectors while ascribing more weight to validated pairs (positive samples) compared to unknown pairs (negative samples). Wang *et al.* [27] propose a DTI prediction method by

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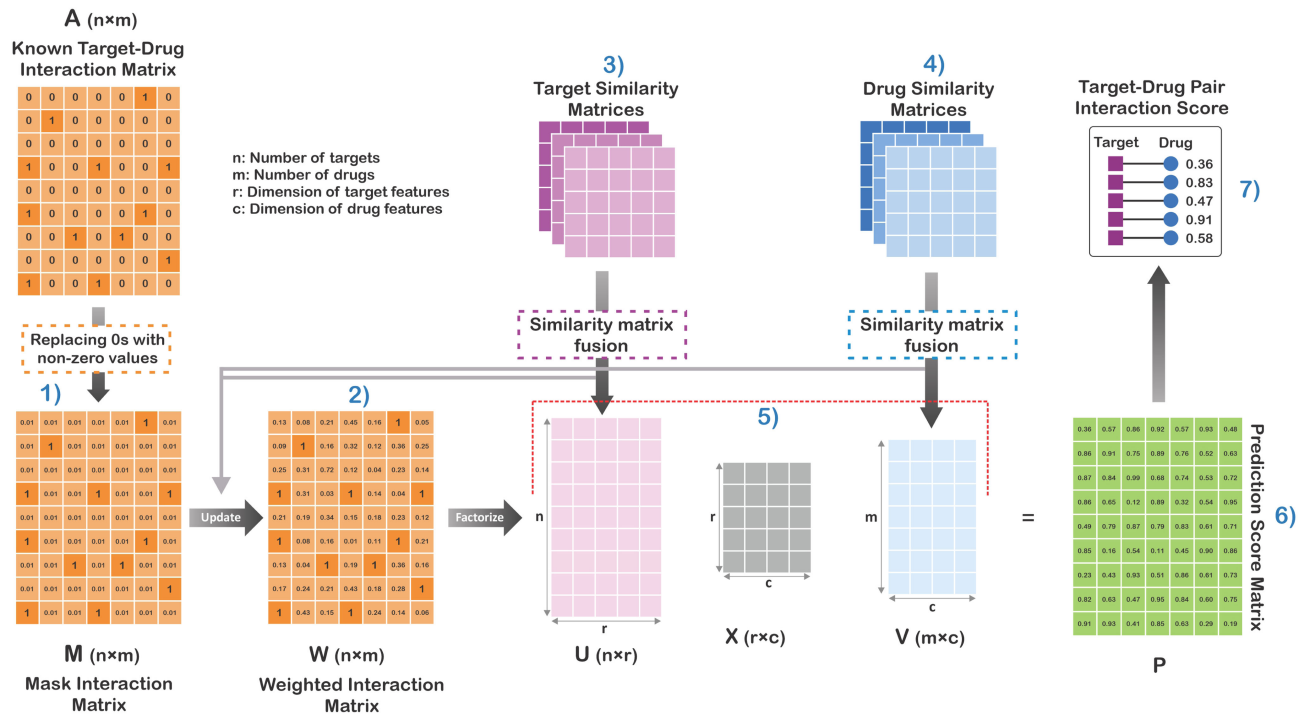


Fig. 1. The workflow of NMTF-DTI. It consists of six steps: 1) replacing zeros with non-zero values in the interaction matrix; 2) weighting the interaction matrix; 3) constructing multiple similarity matrices for targets; 4) constructing multiple similarity matrices for drugs; 5) factorizing the weighted interaction matrix into matrices U , X , and V , and incorporating them with target and drug similarity matrices; 6) constructing the prediction score matrix P ; 7) obtaining an interaction score for each target-drug pair.

integrating diverse heterogeneous data on drugs and targets through Laplacian graph regularized matrix completion. They use this information to construct a similarity graph for drugs and targets to obtain the optimal projection from drug space onto target space. The results of their study confirm that heterogeneous data can significantly improve prediction performance compared with homogeneous networks. You *et al.* [21] use graph theory and a Laplacian-regularized method to combine DTI networks with drug and target similarity networks to construct a heterogeneous network. The results of their experiment confirm that better results can be obtained when various data sources are applied [28].

Although many prediction methods have been developed, there is still room to improve their performance. One key hurdle is overcoming the sparsity of known information on DTIs. The prediction of unknown DTIs requires a method that can extrapolate from existing “hidden” (indirect or implicit) information. Many methods are not able to do this well. Methods with strong generalization ability are better able to make use of such hidden information regarding drugs and targets. Applications of neighbor (similarity) information of drugs and targets accompanied by Laplacian regularization (LR) show high extrapolation ability specifically when there is only a small number of known DTIs and a large number of undiscovered interactions. We hypothesize that the extrapolation capability of LR can significantly improve DTI prediction.

This study proposes a nonnegative tri-factorization method called NMTF-DTI based on Laplacian regularization and the similarity information of drugs and targets to boost the performance of DTI prediction. The contribution of this study is using multiple different similarity measures to a) update the initial interaction matrix and b) to construct a prediction matrix based on the objective function. We also

investigate whether the weighting of the initial interaction matrix can improve the performance of prediction. The novelty of NMTF-DTI comes in the use of a weighted nearest neighbor procedure for updating drug and target interaction profiles, where each drug and target profile is weighted using drug and target similarity information. Another novelty of this work is the use of additional mechanisms to identify which drug similarity measure contributes most to the performance of prediction. Fig. 1 schematically summarizes NMTF-DTI. The results show that our method achieves superior performance compared with competing approaches.

2 MATERIALS AND METHODS

2.1 Data

Four experimentally validated datasets introduced by Yamashita *et al.* [29] are used as gold standard datasets to perform cross-validation in this study: Enzyme (E), Nuclear Receptor (NR), Ion Channel (IC), and G Protein-Coupled Receptor (GPCR). Table 1 summarizes the datasets. Moreover, in order to set parameters, another dataset (dataset S) is used. Dataset S is manually created and consists of recently validated interactions of the above-mentioned four protein

TABLE 1
The Counts of the Entries in the Gold Standard Datasets

	E	IC	GPCR	NR	Total
Proteins	664	204	95	26	989
Drugs	445	210	223	54	932
Interactions	2926	1476	635	90	5127

The count for entries in gold standard datasets; Enzymes (E), Ion Channels (IC), GPCR, Nucleic Receptors (NR).

groups, excluding those pairs existing in the gold standard datasets. The interactions, proteins, and drugs are retrieved from DrugBank [30], UniProt (www.uniprot.org), SuperTarget [31], and KEGG [32]. Dataset S includes 935 proteins, 785 drugs, and 1936 interactions. The similarity measures for drugs are computed using data retrieved from DrugBank [30] and the Comparative Toxicogenomics Database (www.ctdbase.org). The similarity of targets is determined using data retrieved from UniProt (www.uniprot.org), IntAct [33], BIOGRID [34], HPRD [35], STRING [36], MINT [37] and MENTHA [38].

2.2 Problem Formulation

In this study, nonnegative matrix tri-factorization (NMTF) [39] with the Laplacian regularization is used to predict unknown DTIs. The DTI network is represented by a non-negative matrix $A \in \mathbb{R}^{n \times m}$ whose rows (columns) correspond to targets (drugs) where the element (i, j) is 1 if an interaction between target i and drug j is experimentally validated; otherwise, it is 0. The interaction prediction process via NMTF consists of two steps:

Step A) Factorizing the interaction matrix A (or weighted interaction matrix W if the weighting task is performed) into three matrices $U \in \mathbb{R}^{n \times r}$, $X \in \mathbb{R}^{r \times c}$, and $V \in \mathbb{R}^{m \times c}$ where $r, c \leq \min(m, n)$ (Fig. 1, step 5). Compared to traditional bi-factorization, tri-factorization has several advantages. In a bi-factorization, an interaction matrix A is factorized into a product of two factor matrices U and V such that $A = UV^T$, where U and V are referred to as the factor matrices. In such a factorization, the number of columns in U must be the same as that in V so that matrix A is characterized by a sum of outer products between corresponding columns in U and V , which can only capture two types of relations. U and V are related by a one-to-one correspondence (every column in U is related to one column in V). However, the tri-factorization considers $A = UXV^T$ where a factor matrix X allows different numbers of columns in U and V which interact to reconstruct matrix A . In this regard, numerous interactions between column vectors in factor matrices can contribute to an element of the prediction matrix. In this case, the tri-factorization can represent more and more complex types of relations (similar to the complex relations between drugs and targets, genes and diseases, etc.) than a bi-factorization. Furthermore, drug and target feature vectors in a bi-factorization are independent; however, factor matrix X allows drug and target feature vectors to be dependent and provides additional degrees of freedom and more flexibility in incorporating different drug and target relationships to develop a drug/target similarity-based prediction method. Therefore, more relations between drugs and targets can be taken into the account. Moreover, with tri-factorization, the factor matrix X allows different ranges of bases for the factor matrices U and V . Then, the factor matrix X can absorb the scaling effect so that the regularization of matrices U and V can be much easier compared to existing bi-factorization methods. [40], [41], [42], [43], [44].

Step B) Computing new drug-target interaction prediction matrix P (Fig. 1, step 6) as the product of the updated three matrices; i.e., $UXV^T \approx P$.

To solve the problem in step A, the optimization problem with U , X , and V as optimization variables is formulated as follows:

$$\begin{aligned} \arg \min (UXV) &= \frac{1}{2} \| (A - UXV^T) \|_F^2 \\ \text{s.t. } U, X, V &\geq 0 \end{aligned} \quad (1)$$

where $\|Z\|_F^2$ is the Frobenius norm of the matrix Z . Incorporating drug and target similarity matrices into (1) yields the following optimization objective function:

$$\begin{aligned} L(U, X, V) &= \frac{1}{2} \| (A - UXV^T) \|_F^2 + \frac{\lambda_t}{2} \text{Tr}(U^T L_t U) \\ &\quad + \frac{\lambda_d}{2} \text{Tr}(V L_d V^T) \\ \text{s.t. } U, X, V &\geq 0 \end{aligned} \quad (2)$$

where $L_t = D_t - S_t$ ($L_d = D_d - S_d$) is the Laplacian matrix for targets (drugs), S_t (S_d) is the target (drug) similarity matrix described in Sections 2.4 and 2.5, D_t (D_d) is the diagonal matrix whose entries are row sums of S_t (S_d), and λ_t (λ_d) is the regularization parameter to properly integrate the information from target (drug) similarities.

In the initial interaction matrix A , those pairs that are not experimentally validated are filled with 0s. In fact, these pairs potentially could be undiscovered interacting pairs. To take this fact into consideration, first, we replace all zeros with a small non-zero value in the matrix A to construct matrix M (mask matrix). Then, we use the nearest neighbor information of targets (drugs) to construct a weighted interaction matrix W based on the method used by Jamali *et al.* [45] (Fig. 1, step 1). Matrix W is introduced into the objective function to increase the strength of our method. If the matrices A , S_t , and S_d are nonnegative, then matrix W stays non-negative. Moreover, to avoid the risk of overfitting, the Tikhonov regularization term is taken into account and, as a result, the objective function is rewritten as:

$$\begin{aligned} L(U, X, V) &= \frac{1}{2} \| (A \otimes (W - UXV^T)) \|_F^2 \\ &\quad + \frac{\lambda_t}{2} (\|U\|_F^2 + \|V\|_F^2) + \frac{\lambda_t}{2} \text{Tr}(U^T L_t U) \\ &\quad + \frac{\lambda_d}{2} \text{Tr}(V L_d V^T) \\ \text{s.t. } U, X, V &\geq 0 \end{aligned} \quad (3)$$

where $\otimes$ is the Hadamard product of two matrices, and λ_t is a regularization parameter.

2.3 Optimization Process

To construct the three matrices U , X , V , (Fig. 1, step 5) our objective function can be optimized by iterative update rules. To do so, the function is optimized with respect to one matrix while fixing the other two. Then with some mathematical operations, (3) becomes:

$$\begin{aligned} L(U, X, V) &= [\text{Tr}((A \otimes W)^T (A \otimes W) \\ &\quad - 2(A \otimes W)^T (A \otimes (UXV^T)) + (A \otimes (UXV^T))^T \\ &\quad + (A \otimes (UXV^T)))] + \frac{\lambda_t}{2} \text{Tr}(U^T L_t U) + \frac{\lambda_t}{2} \text{Tr}(UU^T) \\ &\quad + \frac{\lambda_d}{2} \text{Tr}(V L_d V^T) + \frac{\lambda_t}{2} \text{Tr}(VV^T) \\ \text{s.t. } U, X, V &\geq 0 \end{aligned} \quad (4)$$

The partial derivatives of the above equation with respect to U , X , and V are, respectively:

$$\begin{aligned}\frac{\partial L(UXV)}{\partial U} &= -(A \otimes W)(VX^T) + (A \otimes (UXV^T))(VX^T) \\ &\quad + \lambda_t L_t U + \lambda_l U \\ \frac{\partial L(UXV)}{\partial X} &= -U^T(A \otimes W)^T V + U^T(A \otimes (UXV^T))V \\ \frac{\partial L(UXV)}{\partial V} &= -(A \otimes W)^T(UX) + (A^T \otimes (VX^T U^T))(UX) \\ &\quad + \lambda_d L_d V + \lambda_l V\end{aligned}\quad (5)$$

The Karush-Kuhn-Tucker (KKT) conditions are used to ensure the non-negativity of U , X , V yielding:

$$\begin{aligned}[-(A \otimes W)(VX^T) + (A \otimes (UXV^T))(VX^T) + \lambda_t D_t U \\ + \lambda_t S_t U + \lambda_l U]_{ij} u_{ij} &= 0 \\ [-U^T(A \otimes W)^T V + U^T(A \otimes (UXV^T))V]_{ij} x_{ij} &= 0 \\ [-(A \otimes W)^T(UX) + (A^T \otimes (VX^T U^T))(UX) + \lambda_d D_d V \\ + \lambda_d S_d V + \lambda_l V]_{ij} v_{ij} &= 0\end{aligned}\quad (6)$$

With some modification, we arrive at the following update rules [46]:

$$u_{ij} \leftarrow u_{ij} \sqrt{\frac{[(A \otimes W)(VX^T) + \lambda_t S_t U]_{ij}}{[A \otimes (UXV^T)](VX^T) + \lambda_t D_t U + \lambda_l U]_{ij}}}\quad (7)$$

$$x_{ij} \leftarrow x_{ij} \sqrt{\frac{[U^T(A \otimes W)^T V]_{ij}}{[U^T(A \otimes (UXV^T))V]_{ij}}}\quad (8)$$

$$v_{ij} \leftarrow v_{ij} \sqrt{\frac{[(A \otimes W)^T(UX) + \lambda_d S_d V]_{ij}}{[(A^T \otimes (VX^T U^T))(UX) + \lambda_d D_d U + \lambda_l V]_{ij}}}\quad (9)$$

Upon convergence, the prediction matrix for drugs and targets is obtained as $P \approx UXV^T$ where the element (i, j) represents the interaction score of target i and drug j (Fig. 1, step 7).

2.4 Drug Similarity Matrices

Most DTI prediction methods leverage the assumption that if drug D interacts with a target, then drugs similar to drug D can interact with the same target and vice versa [47]. However, using a single type of similarity measure may miss relevant information in predicting new interactions. Hence different similarity measures utilize different attributes of drugs or targets and calculate their similarities from different perspectives. To increase the performance of prediction, multiple similarity measures for drugs and targets are calculated based on different chemical and biological sources as follows:

Chemical structure-based similarity: Most of the structure-based methods rely on the simplified molecular input line entry system (SMILES) to calculate chemical compound and drug similarity. SMILES strings of approved drugs are retrieved from DrugBank to construct a drug similarity matrix using the ChemmineOB and ChemmineR [48] packages in the R programming language. Different measures

can be used to compute the similarity of compounds. The Tanimoto measure is used to calculate the similarity between two drugs d_i and d_j based on the two-dimensional structure as $S_{Chem}(d_i, d_j)$.

Annotation-based similarity: The anatomical, therapeutic, and chemical (ATC) classification system of drugs is widely used to measure drug similarity. Drugs are categorized at five levels according to their hierarchical structure and classification. We obtained the ATC codes of approved drugs from DrugBank. These codes are used to calculate the similarity of drugs using the semantic similarity measure for each level. Then the average value of five levels $S_{ATC}(d_i, d_j)$ is considered as the similarity of drugs d_i and d_j .

Target similarity-based similarity: It is assumed that similar drugs can interact with similar targets. Based on this assumption, the similarity of drugs is calculated based on their interacting targets. The semantic similarity of targets is computed according to the gene ontology (GO) of targets. Then the similarity of two drugs (d_i, d_j) is calculated by integrating the target similarity of drugs as follows:

$$S_{Tar}(d_i, d_j) = \frac{\sum_{m=1}^{N_i} \sum_{n=1}^{N_j} S_{GO}(t_i, t_j)}{N_i \times N_j}\quad (10)$$

where $S_{GO}(t_i, t_j)$ is the semantic similarity of targets t_i and t_j described in the next Section, N_i (N_j) is the total number of targets interacting with drug d_i (d_j).

Drug-disease association-based similarity: Based on the assumption that similar drugs can be used in similar diseases, drug similarity as S_{Dis} is calculated using the associations between drugs and diseases. The semantic similarity measure of diseases is used to measure the similarity of drugs d_i and d_j . First, the similarity of diseases D_i and D_j is calculated according to the method of Zhang *et al.* [49]. Then the similarity of two drugs (d_i, d_j) is computed as follows:

$$S_{Dis}(d_i, d_j) = \frac{\sum_{m=1}^{N_i} \sum_{n=1}^{N_j} S(D_i, D_j)}{N_i \times N_j}\quad (11)$$

where $S_{Dis}(D_i, D_j)$ is the similarity of disease D_i and D_j , N_i and N_j are the total number of diseases associating with drugs d_i and d_j , respectively.

2.5 Target Similarity Matrices

Sequence-based similarity: The structure information of targets is used to compute the sequence-based similarity. We download the structure and sequence of targets from UniProt [50]. Then a normalized version of the Smith-Waterman algorithm (12) is used to calculate the score between sequences of each pair of targets. The normalized Smith-Waterman score between two targets t_i and t_j is computed as:

$$S_{Seq}(t_i, t_j) = \frac{SW(t_i, t_j)}{\sqrt{SW(t_i, t_i)SW(t_j, t_j)}}\quad (12)$$

where $SW(t_i, t_j)$ denotes the original Smith-Waterman score between two targets t_i and t_j . Then the matrix of sequence similarities is constructed for protein sequences as $S_{Seq}(t_i, t_j)$.

Protein-protein interaction (PPI) network similarity: Human PPI networks are retrieved from the UniProt, IntAct [33], BioGRID [34], HPRD [35], STRING [36], MINT [37] and Mentha [38] databases. First, the duplicates are removed, and then the similarity of all pairs of those targets, which are presented in the above-mentioned PPI networks, is calculated based on the shortest-path distance between the targets as follows:

$$S_{PPI}(t_i, t_j) = Ae^{-bD(t_i, t_j)} \quad (13)$$

where $D(t_i, t_j)$ is the shortest path between targets t_i and t_j in the PPI network, A and b are free parameters which are set to 0.9 and 1, respectively.

Gene Ontology (GO) semantic similarity: All three categories of GO annotations for targets are obtained from UniProt. Then, the semantic similarity between GO terms of targets is computed using GOSim [51] and GOSemSim [52] packages in the R programming language as $S_{GO}(t_i, t_j)$.

Finally, the various similarity matrices for drugs and targets are combined based on the Similarity Network Fusion (SNF) method developed by Wang *et al.* [53] to construct the final fused similarity matrix for drugs and targets (S_t and S_d) (Fig. 1, steps 3 and 4). Also, to identify the significance of each drug similarity measure in the prediction, we systematically combine three drug similarity matrices while excluding one and determined the performance of the resulting model. See Supplementary Materials, which can be found on the Computer Society Digital Library at <http://doi.ieeecomputersociety.org/10.1109/TCBB.2021.3135978>. Section 1 to find out how similarity matrices are combined.

2.6 Evaluation

Two cross-validations (CV) with different datasets are performed in this study: CV for setting parameters and CV for performance evaluation. The cross-validation needs two types of data samples: positive and negative samples. Verified interactions are used as positive samples. Generally, non-interacting pairs are used as negative samples; however, these pairs can be potentially true interactions that are not yet verified. To overcome this issue, we filter non-interacting pairs using the longest shortest path based on the method developed by Lou *et al.* [54] to remove potentially interacting pairs (See Supplementary Materials, Section 2, available online). Then an equal number to the positive samples are randomly chosen from non-interacting pairs. Finally, all positive and negative samples are randomly divided into 10 equal subsets. Each subset is used in turn as a test set and the remaining subsets are used as training sets. As the results of cross-validation, AUC and AUPR are calculated, and the best performance is chosen based on these metrics.

2.6.1 Cross-Validation for Parameters Setting

Parameter setting is performed using cross-validation utilizing the dataset S described in Section 2.1. Different sets of parameters are considered in our objective function including: k , the neighborhood size from 5 to 15 with step size 1; r , the rank of matrix U from {100, 150, 200}; c , the rank of matrix V from {50, 100}; λ_t and λ_d from {0, 0.000001, 0.00001, 0.0001, 0.001, 0.01, 0.1}; and λ_l from {0, 0.1, 0.2, 0.3, 0.4, 0.5}. To select the optimum set of the above parameters, 10-fold cross-

validation is performed for all combinations of parameters. Finally, the best combination of parameters is chosen based on the best value for AUC.

2.6.2 Cross-validation for Performance Evaluation

To evaluate the performance of our method, cross-validation is performed in three scenarios as follows:

- 1) CV_{drug-target}: cross-validation on the drug-target pairs, which aims to predict new (undiscovered) drug-target pairs.
- 2) CV_{target}: cross-validation on targets which tries to predict new targets for the existing drug(s).
- 3) CV_{drug}: cross-validation on drugs which refers to the prediction of new drugs for existing target(s).

Moreover, the performance of our method is compared with some state-of-the-art algorithms for the DTI prediction.

2.7 Case Study

In order to study the reliability of NMTF-DTI, we randomly select one drug from each dataset within the gold standard datasets, remove that drug's profile while performing CV_{drug}, and observe the top 10 predicted interactions made by the different methods with targets within the corresponding dataset.

3 RESULTS

In this section, we describe the experimental results of our proposed method. We first use CV to set the parameters. Then, we use CV to evaluate if weighted matrix can improve prediction performance (See Supplementary Materials, available online, Section 3). Moreover, the performance of NMTF-DTI in terms of AUC and AUPR is compared to that of existing state-of-the-art methods, including NRLMF [23], NRLMF β [55], FRnet-DTI [56], BICTR [7], and a non-negative matrix tri-factorization method [57] that we have named SNMTF in this paper, run with the same input data and same CV settings. Note that NRLMF β is a modified version of NRLMF algorithm where it is improved with beta distribution rescaling. In addition, the result of the case study is described.

3.1 Implementation

Our proposed method and comparison approaches are implemented using the Scikit-learn [58], Numpy [59], PyBioMed [60], and Biopython [61] libraries in Python language (version 3.6). The R programming language with packages ChemmineOB and ChemmineR are used to calculate drug similarity.

3.2 Parameter Sensitivity Analysis

As discussed in Section 2.6.1, various parameters need to be tuned to obtain the best performance and results. We perform a CV on a dataset S with all combinations of these parameters.

The optimal value for each parameter is selected based on the AUC as follows. The results confirm that increasing numbers of neighbors (k) significantly increases the performance of prediction until $k = 11$, after which larger values of k do not affect the performance of prediction. The best result is achieved by setting $\lambda_t = \lambda_d = 0.0001$, $\lambda_l = 0.2$, and $r = 150$ and $c = 50$ for the rank of matrices U and V , respectively.

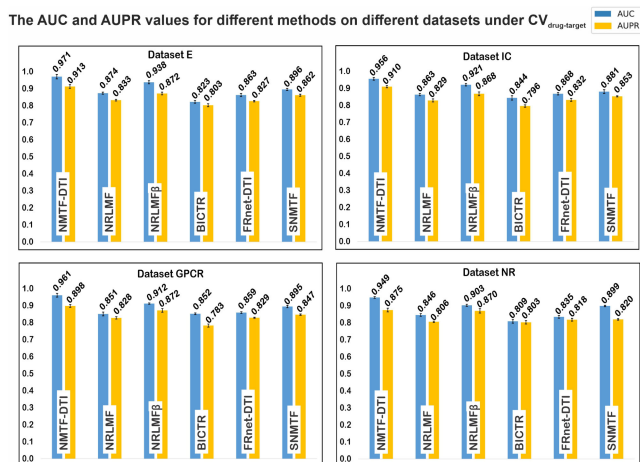


Fig. 2. The AUC and AUPR values for the compared methods on four benchmark datasets under the CV_{drug-target} scenario.

3.3 Evaluation Metrics

To evaluate the performance of our method, the experiments are conducted on the gold standard datasets described in Section 2.1. Similar to the evaluation of most DTI prediction methods, AUC and AUPR are used as the main metrics for performance evaluation. The contribution of each drug similarity matrix in performance of NMTF-DTI is evaluated. Supplementary Tables S1–S3, available online, shows that the best result is achieved when all four drug similarity matrices are combined. Furthermore, Supplementary Tables S4–S6, available online, confirm that using a weighted matrix improves the performance of prediction. In addition, in terms of comparison, our method outperforms compared to the other methods. Figs. 2, 3, and 4 and Supplementary Tables S7–S9, available online, show the AUC and AUPR values for different prediction methods on different datasets under the three CV scenarios. As can be seen, our proposed NMTF-DTI performs better than the other competing methods on all datasets.

Fig. 2 shows the AUC and AUPR values for all competing methods under the CV_{drug-target} scenario. As shown, our method obtains the best AUC among all competing methods for datasets E (0.971), IC (0.956), GPCR (0.961), and NR

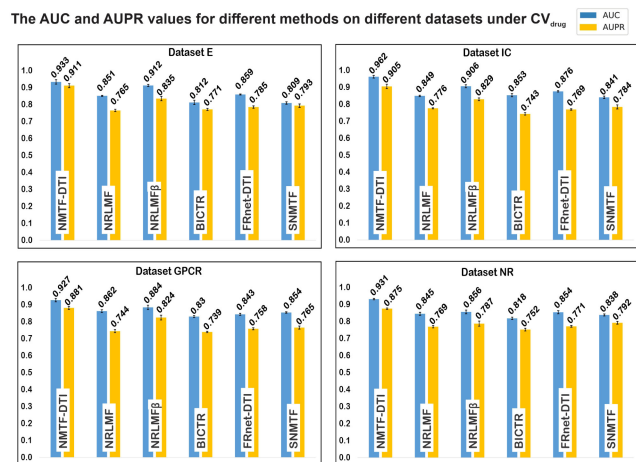


Fig. 4. The AUC and AUPR values for the compared methods on four benchmark datasets under the CV_{drug} scenario.

(0.949). In addition, NMTF-DTI obtains the best AUPR for E (0.913), IC (0.910), GPCR (0.898), and NR (0.875) datasets among all methods. Moreover, the average values for AUC (0.959) and AUPR (0.899) are the best among all competing methods. The results suggest that the performance of NMTF-DTI is better compared with those methods using a single similarity measure (i.e., NRLMF, NRLMFβ).

The results obtained under the CV_{target} scenario are summarized in Fig. 3. As summarized, NMTF-DTI attains the best AUC and AUPR on all datasets. NMTF-DTI achieves AUC values of 0.941, 0.953, 0.961, and 0.954 and values of 0.908, 0.918, 0.9920, and 0.915 for AUPR on the E, IC, GPCR, and NR datasets, respectively. NMTF-DTI obtains the best average values for AUC (0.952) and AUPR (0.915) on all datasets. The average AUC and AUPR obtained by NMTF-DTI are significantly higher than those achieved by the second-best method.

Furthermore, Fig. 4 represents the results obtained under the CV_{drug} scenario. Similar to the other two CV scenarios, NMTF-DTI achieves the best results for AUC and AUPR on all datasets among all competing methods. NMTF-DTI obtains the highest AUC value for datasets E (0.933), IC (0.962), GPCR (0.927), and NR (0.931). Moreover, our method obtains 0.911, 0.905, 0.881, and 0.875 for AUPR on E, IC, GPCR, and NR datasets, respectively. NMTF-DTI obtains the best average values for AUC (0.940) and AUPR (0.893) on all datasets.

In general, NMTF-DTI can integrate drug and target information from multiple sources and show better results compared with the results of those methods using single similarity matrices for drugs and targets.

3.4 Case Study

The results of the case study are presented in Tables 2, 3, 4, and 5 and Fig. 5. As shown in Table 2, Methoxsalen is selected as the target drug from the Enzyme (E) dataset. For this dataset, our method predicts 10 verified interacting targets as its top 10 predictions while other methods predict fewer. The result supports the conclusion that NMTF-DTI can predict the most verified interactions compared to the other prediction methods on this dataset. NRLMFβ is the second-best method on the Enzyme dataset,

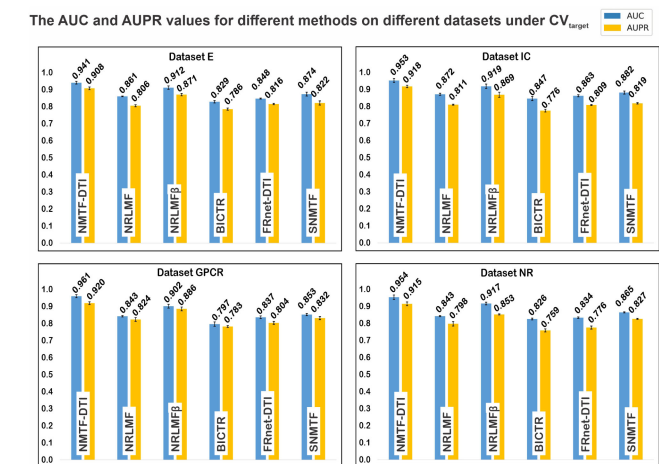


Fig. 3. The AUC and AUPR values for the compared methods on four benchmark datasets under the CV_{target} scenario.

TABLE 2

The Top 10 Interacting Targets of Drug Methoxsalen in the Enzymes Dataset Predicted By the Different Methods

	NMTF-DTI	NRLMF	NRLMF β	BICTR	FRnet-DTI	SNMTF
1	1593	1551	5740	1595	9023	10935
2	5740	2322	1562	1553	1675	9420
3	64816	1582	10667	10924	5740	29785
4	1584	109	1577	1147	660	1562
5	10858	4051	109	1581	1582	108
6	9023	5138	1595	9420	1580	10858
7	29785	1594	1551	1021	3643	5740
8	1555	3067	9023	5740	84171	2263
9	1582	1573	1360	10720	199974	1557
10	4051	1580	1593	2045	1548	1553

The entries represent the number portion of human protein ID in KEGG database (i.e., 1593 means hsa1593). Experimentally verified targets are indicated in **boldface** and underlined. The number of experimentally verified targets in the drug profile is 43.

TABLE 3

The Top 10 Interacting Targets of Drug Nisoldipine in the Ion Channels Dataset Predicted By the Different Methods

	NMTF-DTI	NRLMF	NRLMF β	BICTR	FRnet-DTI	SNMTF
1	59284	6261	8912	5310	6262	55503
2	3710	3747	775	55799	3710	9254
3	93589	1181	5310	1139	200909	8912
4	8913	777	3757	2895	777	5310
5	781	774	6261	785	6324	56479
6	6263	55879	2554	2555	10369	8913
7	3708	8911	8912	3708	8989	3787
8	55503	2554	777	3766	3359	786
9	785	27092	3710	23630	59284	3708
10	6261	8913	8989	8989	2555	2561

The entries represent the number portion of human protein ID in KEGG database (i.e., 59284 means hsa59284). Experimentally verified targets are indicated in **boldface** and underlined. The number of experimentally verified targets in the drug profile is 30.

predicting seven verified interacting targets out of its top 10 predictions.

From the IC dataset, Nitrendipine is randomly chosen. Table 3 summarizes the top 10 interacting targets of the drug Nisoldipine in the IC dataset predicted by each method. As shown, all 10 of NMTF-DTI's predictions are verified interaction targets. For this dataset, NRLMF β has the second-most verified interaction targets (eight) amongst its top 10 predictions.

For dataset GPCR, Pindolol is randomly selected as the subject drug. Table 4 shows that among all methods, NMTF-DTI is the best method where, of its 10 predicted interaction targets, eight are verified ones. For this dataset, NRLMF β is the second-best method, predicting seven verified interacting targets out of its top 10.

Table 5 summarizes the top 10 interactions between Tazarotene and targets in the NR dataset. As can be seen in the table, our proposed method predicts the most verified interaction targets (seven). As the second-best methods, NRLMF β predicts six verified interaction targets.

The results provided in Tables 2, 3, 4, and 5 confirm that our method is the most reliable in the prediction of interactions between targets and drugs. Fig. 5 summarizes the fraction of experimentally verified interactions in the top 10

TABLE 4

The Top 10 Interacting Targets of Drug Pindolol in the GPCR Dataset Predicted By the Different Methods

	NMTF-DTI	NRLMF	NRLMF β	BICTR	FRnet-DTI	SNMTF
1	3351	5028	3355	3351	3358	154
2	3356	151	3362	6915	3274	2911
3	3362	152	5724	2912	153	3358
4	153	3352	151	3360	3362	5729
5	150	4988	2918	5737	5032	6010
6	2550	3350	3350	552	3352	3355
7	3358	1909	3357	151	4988	1132
8	554	2915	135	1815	1909	3360
9	3354	3360	3354	146	1813	3350
10	3350	3361	147	3362	155	151

The entries represent the number portion of human protein ID in KEGG database (i.e., 3351 means hsa3351). Experimentally verified targets are indicated in **boldface** and underlined. The number of experimentally verified targets in the drug profile is 19.

TABLE 5

The Top 10 Interacting Targets of Drug Tazarotene in the Nuclear Receptor Dataset Predicted By the Different Methods

	NMTF-DTI	NRLMF	NRLMF β	BICTR	FRnet-DTI	SNMTF
1	6257	5465	5916	7421	6258	7421
2	5915	3174	2104	9971	5241	5914
3	5914	5916	6256	5916	6096	5241
4	3174	2101	7421	367	8856	9971
5	6096	190	6095	3174	2100	5915
6	6256	6256	5914	5914	6257	2104
7	5241	2908	9971	2101	6095	6256
8	5916	6258	6257	5468	3174	2908
9	6258	8856	6258	6257	5914	6096
10	7421	5914	5915	2099	4306	6257

The entries represent the number portion of human protein ID in KEGG database (i.e., 6257 means hsa6257). Experimentally verified targets are indicated in **boldface** and underlined. The number of experimentally verified targets in the drug profile is 7.

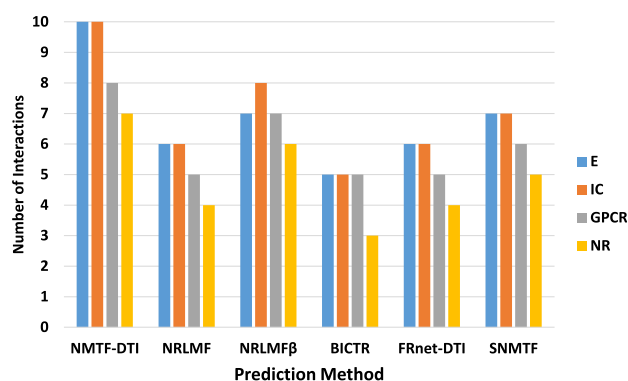


Fig. 5. The fraction of verified interaction targets in the top 10 predicted targets from the different methods starting with the drugs randomly chosen from the four datasets.

predicted interactions made by the different methods starting with each of four randomly selected drugs.

4 CONCLUSION

This study has presented NMTF-DTI as a novel approach to predict interactions between targets and drugs. The novelty

of our method comes from using neighbor (similarity) information on drugs and targets to construct a weighted interaction matrix to overcome the negative impact of undiscovered interactions in the interaction matrix. The other contribution of our model is using multiple similarity matrices, including structural and functional measures for both targets and drugs, and integrating these matrices with the interaction matrix using a matrix tri-factorization approach to predict the interaction score of a given target-drug pair.

To evaluate the performance of NMTF-DTI, a wide set of experiments are carried out on four datasets. To study either the significance and generalization performance of our method, three scenarios ($CV_{\text{drug-target}}$, CV_{target} , and CV_{drug}) are used on four experimentally validated datasets (Enzyme (E), Nuclear receptor (NR), Ion channel (IC), and the G protein-coupled receptor (GPCR)) from the literature [29]. NMTF-DTI is compared with other state-of-the-art methods under the three CV scenarios in terms of AUC and AUPR. The results confirm the superiority of our method on E, IC, GPCR, and NR datasets under the $CV_{\text{drug-target}}$, CV_{target} , and CV_{drug} scenarios. The results of comparing the performance of NMTF-DTI with related methods, those that often use only one type of similarity information, suggests that the construction of target and drug similarity matrices is a critical step for a prediction system, and multiple similarity measures can boost the performance of prediction. Having discussed all advantages of NMTF-DTI over the all methods, it is worthy to mention the limitations of this method. Compared with traditional bi-factorization methods, tri-factorization algorithms take longer running time. Tri-factorization algorithms would be more widely used if their time complexity was more thoroughly studied and improved.

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